GMI German-Malaysian Institute

DIPLOMA PROGRAMME

PBL ASSIGNMENT

Academic Period : Jan 2025

Code	SPG 0473	Course Name	WEB PROGRAMMING
Title	DIAMOND RENTAL	Examiner	M. LUQMAN HAKIM SABANI
Name (Individual /group)	• AIMAN DANISH BIN MOHD KHAIRUNNIZAM *(CBS24070471)* • ABDUL HARIZ BIN MOHD SARNUDIN *(CBS24070208)* • MUHAMMAD RYAN BIN HIFZI *(CBS24070151)* • MUHAMMAD FARIS BIN MOHD FAUZI *(CBS24070213)*	Program/ Group	CBS 1 GROUP 5

No.	Assessment Criteria	CLO	PO	Level of diff	Full Marks	Score	Marks	Comment
1.	Login page with validation	2	4	P3	10			
2.	HTML form with all input type	2	4	P3	10			
3.	Add all buttons needed	2	4	P3	40			
4.	All pages navigated	2	4	P3	10			
5.	Presentation	3	8	A3	10			
6.	Format Report	3	8	P3	10			
7.	Plagiarism	3	8	A3	10			

Total					**100**			
Total Marks								
				CLO 2	70			
				CLO 3	30			
Signature								

PBL ASSIGNMENT QUESTION

You are required to develop an interactive website and PBL Assignment report that must include the following list of criteria:

- Add a login page with validation before entering the system.
- Add an HTML form that allows users to enter input using HTML input types such as text fields, text areas, dropdown menus, checkboxes or any other input type.
- Add any appropriate button such as submit, search, view, delete or any other button.
- All pages must be properly navigated.
- Add any appropriate features to make the system more flexible and interactive such as hyperlinks or any other HTML elements.

Choose one of the following titles (or select another title):

- Library System
- Inventory System
- Ticketing System
- Student Registration System
- Other Topic (specific)

Report (Table of Content)

Introduction (main function, targeted user… students, cars, etc, location)

Problem Statement

Objective

Screenshot all outputs (Explain briefly)

Conclusion

References

FORMAT OF REPORT

- Font type : Times New Roman/ Arial
- Font Size : 12
- Line Spacing : 1.5/ 2.0
- Alignment : Justify
- Presentation & Demonstration
    - Each student must participate in presentation session
    - Submit Report and slide thru Google Drive Link.
- Submission Date: 26/09/2025

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Introduction

Our PBL project is about developing a car rental website called Diamond Rental, created using HTML. The main purpose of this website is to provide a simple and user-friendly platform where users can browse available vehicles and make bookings online. It is designed with students and young adults in mind, especially those who may not own a vehicle but still need transport for studies, daily use, or leisure.

What makes Diamond Rental unique is the variety of vehicles offered. The platform does not only feature practical and affordable cars, but also includes supercars and superbikes for those looking for exciting, premium experiences. This allows the service to meet both everyday needs and special occasions.

The project focuses on local urban areas, particularly around colleges and universities, where affordable, flexible, and convenient transport options are highly valuable. Through this project, we aim to demonstrate how basic web technologies like HTML can be applied to design a functional and engaging platform that responds to real human needs.

Problem Statement

In this project, we were required to develop an interactive website that demonstrates our understanding of HTML and PHP. The system must include a secure login page with validation, an HTML form that uses various input types such as text fields, checkboxes, radio buttons, dropdown menu. and appropriate buttons to improve user interaction such as submit and navigation buttons. All pages must be well connected and navigable and the system should include additional features to make it more flexible and interactive, such as hyperlinks, dynamic content, and a structured layout.

Objective

- Develop an interactive rental website that showcases luxury cars and superbikes with a modern and professional design.

- Implement secure user authentication and session management so only registered users can access the system.

- Integrate interactive features such as HTML forms, a dynamic slideshow, and well-structured navigation for an improved user experience.

- Provide a seamless rental and checkout process including a shopping cart, cost calculation, and downloadable PDF receipts.

- Enhance user engagement and admin communication by adding a feedback submission system that sends feedback to the administrator’s email.

Screenshot all outputs

Accessing the page without an account will automatically redirect the user to the login page. Users who don't have an account can click the register button to create one before renting a car.

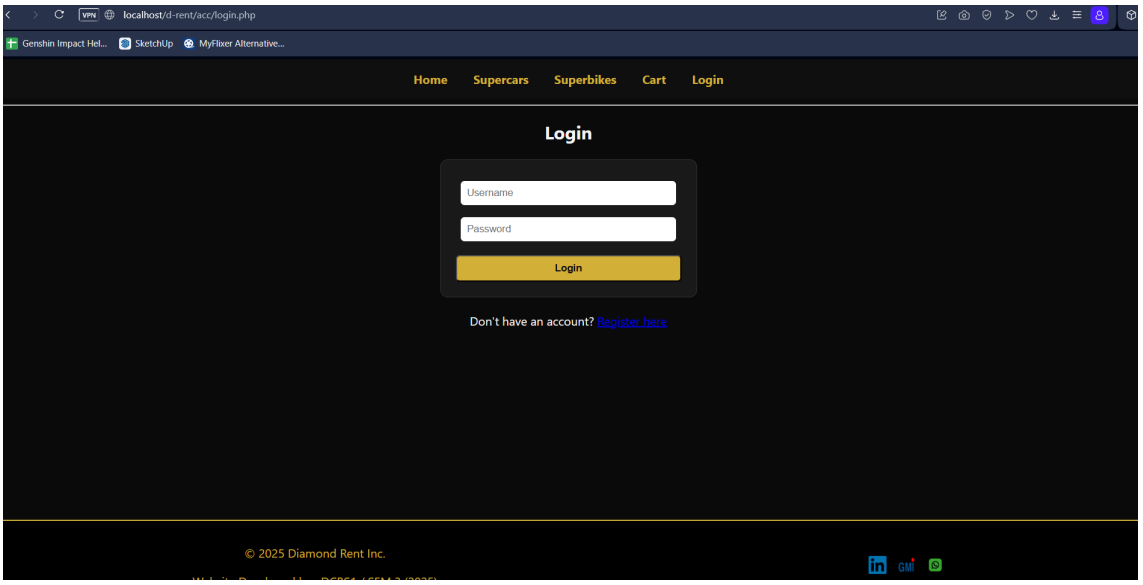


Figure 1.0 : Login Page

If the user is already logged in, clicking the 'Log Out' button will sign them out and automatically redirect them to the login page.

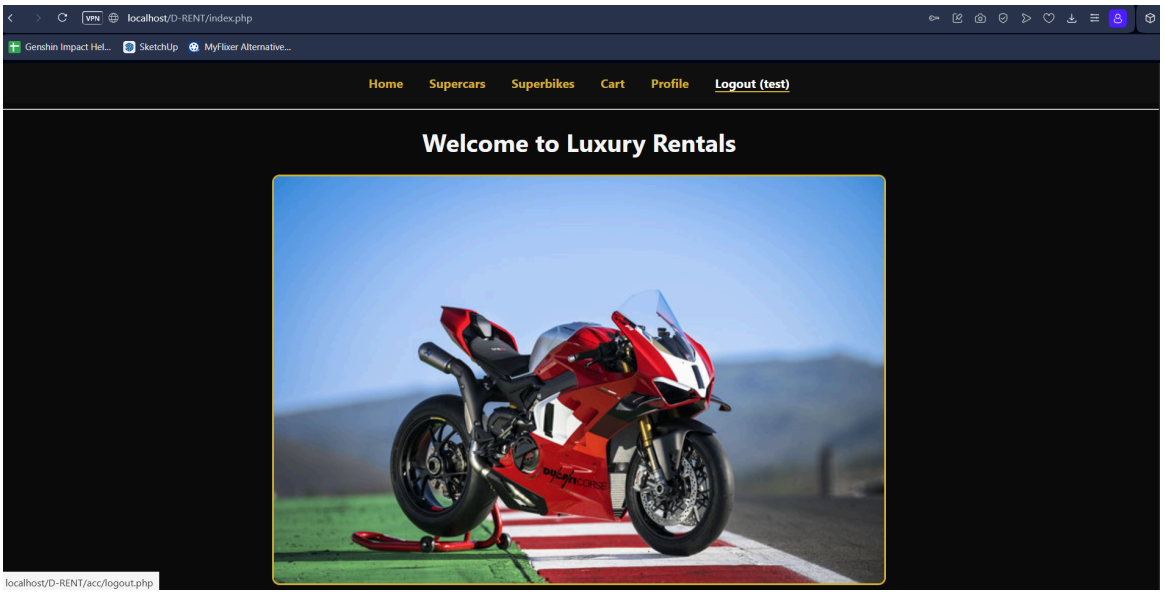


Figure 1.1 : Homepage & Logout Function

Users can create a new account through the register page. The account information is saved in user.txt (a simple mockup database) on the server, allowing them to log in later to rent cars.

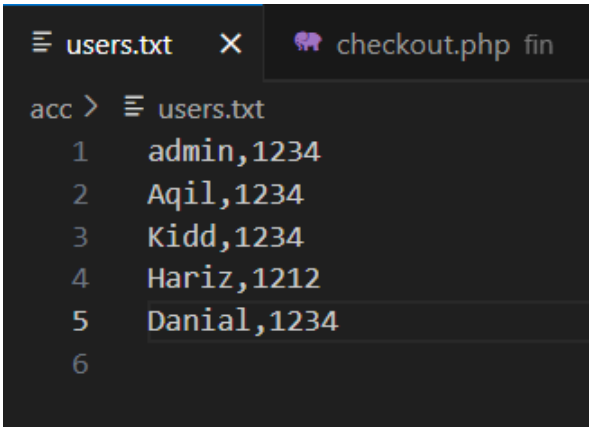


Figure 1.2 : users.txt - mockup db

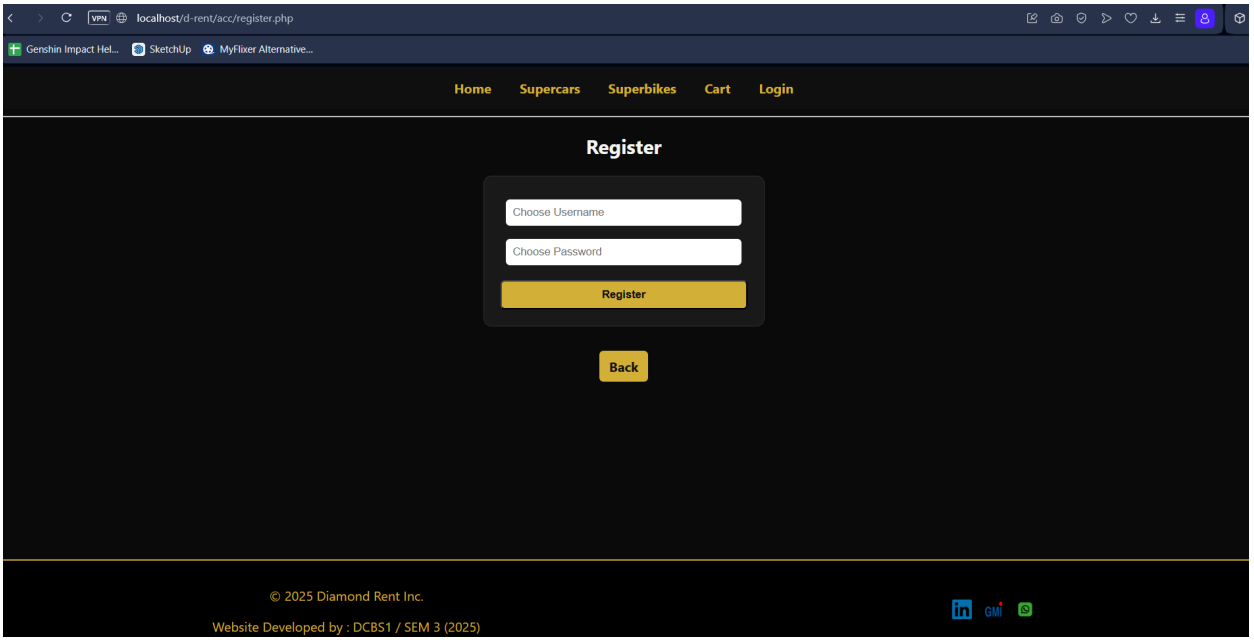


Figure 1.3 : Register Page

After logging in, users are automatically redirected to the main menu (index.php), where they can view and choose from the available cars and bikes.

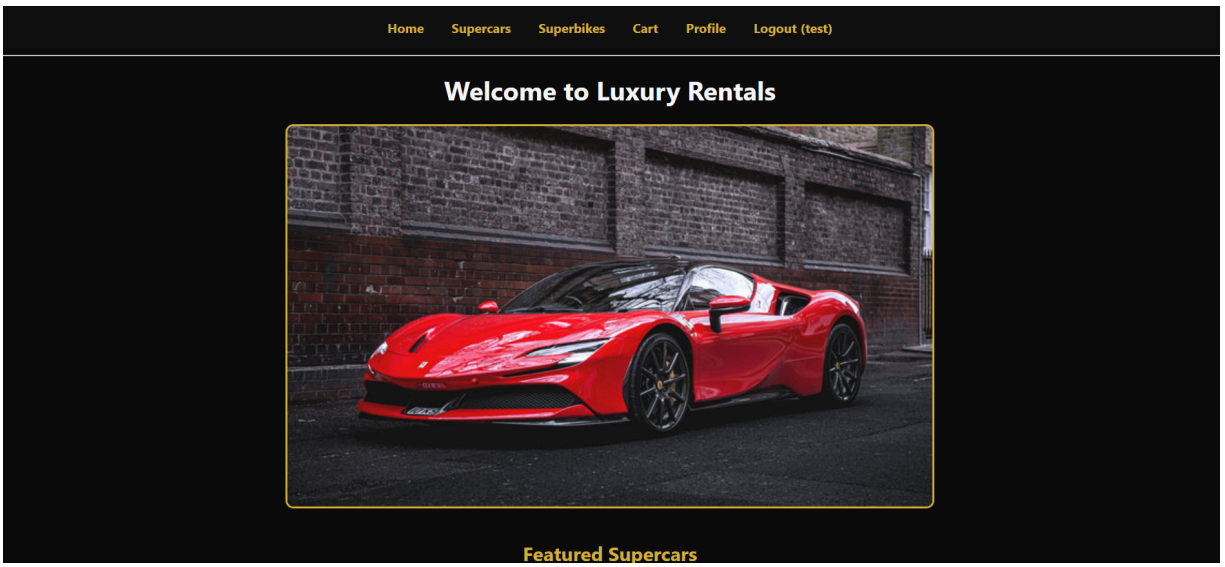


Figure 1.4 : Homepage (index.php)

The home page displays featured cars and bikes, while the dedicated 'Cars' and 'Superbikes' pages offer a wider selection of vehicles for users to browse.

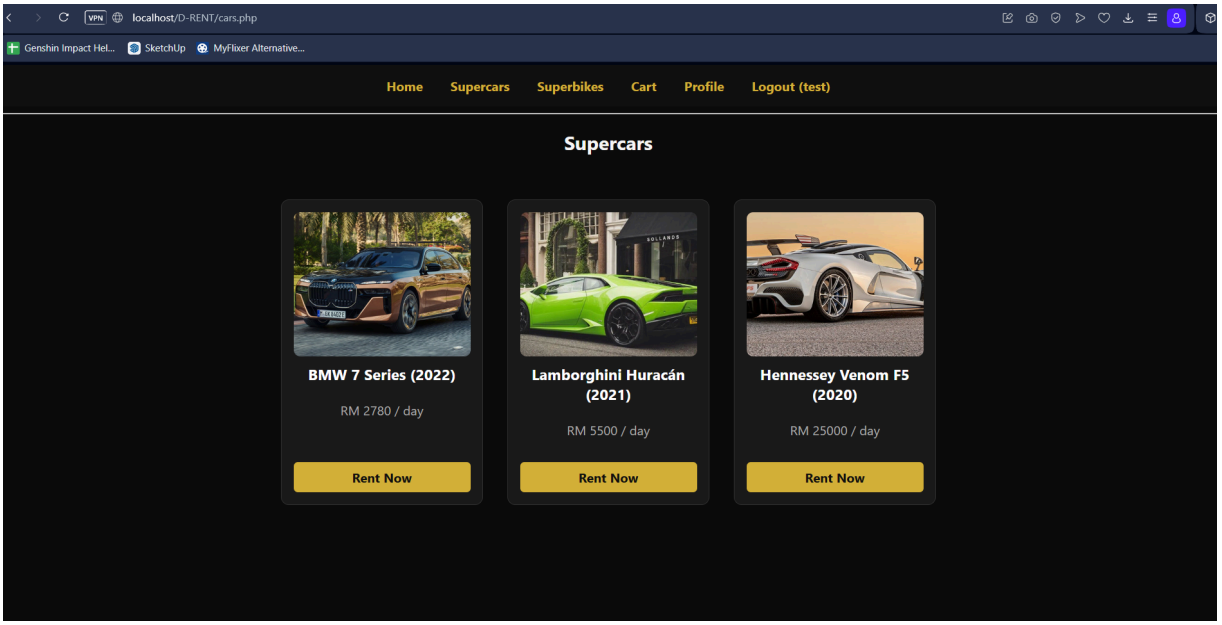


Figure 1.5 : Supercars page

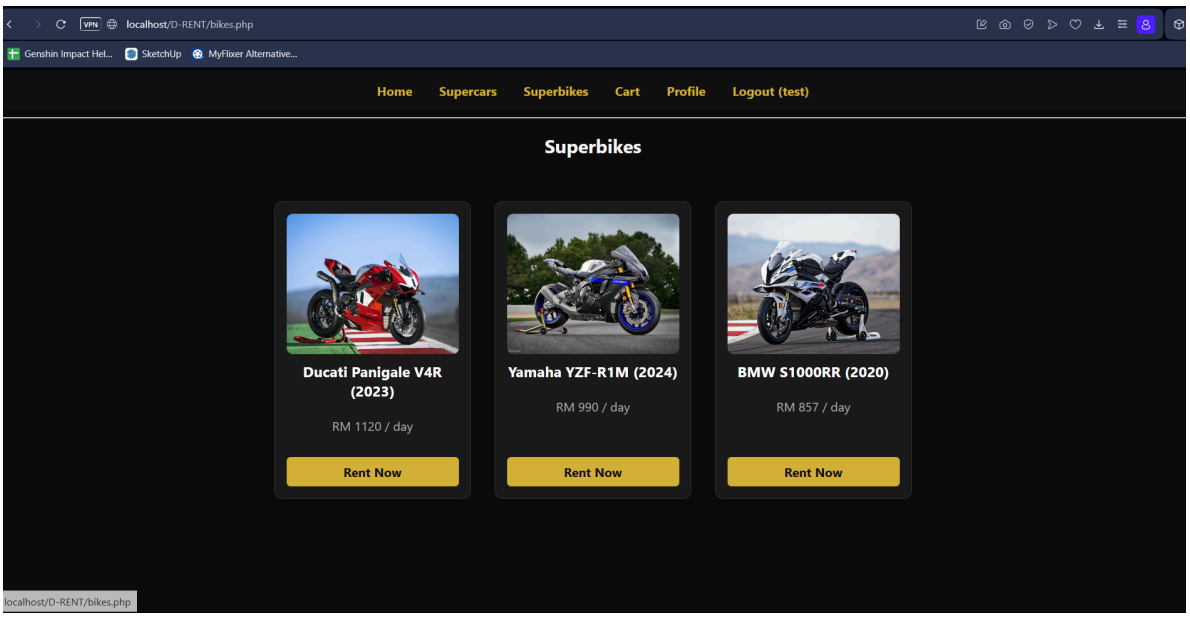


Figure 1.6 : Bikes page

When a user selects a vehicle, for example the Yamaha R1, and clicks the 'Rent Now' button, it is immediately added to their cart for checkout.

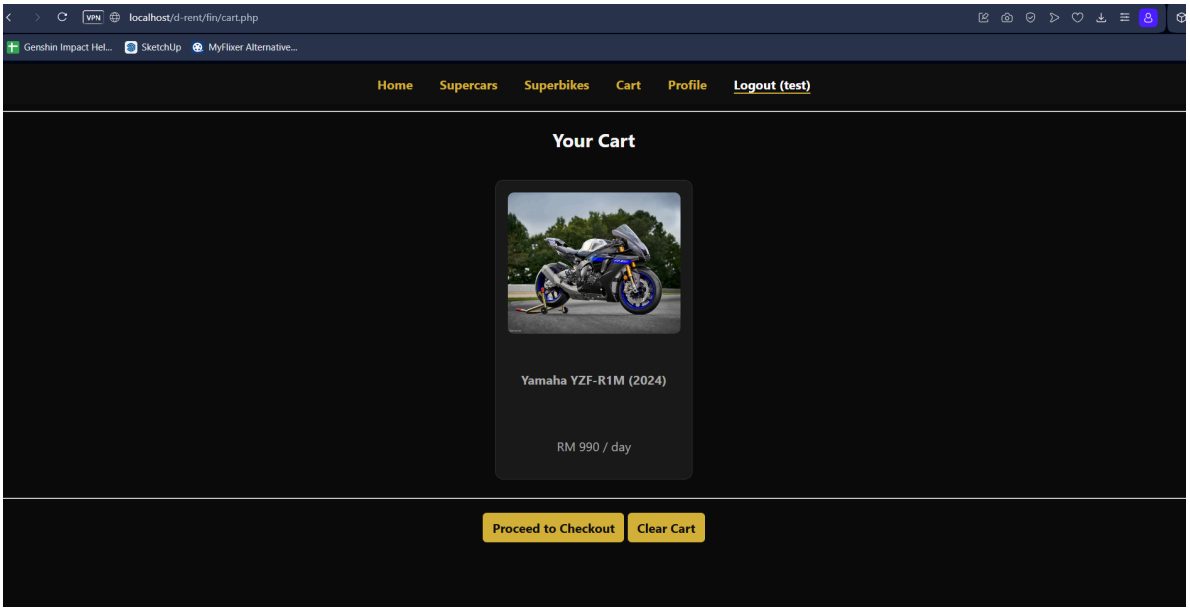


Figure 1.7 : Cart

The Clear Cart system allows users to remove all items from their cart in a single action. By clicking the 'Clear Cart' button, every rented item currently in the cart is deleted, giving the user an empty cart to start fresh. This helps prevent accidental rentals and makes it easier to manage selections before checkout.

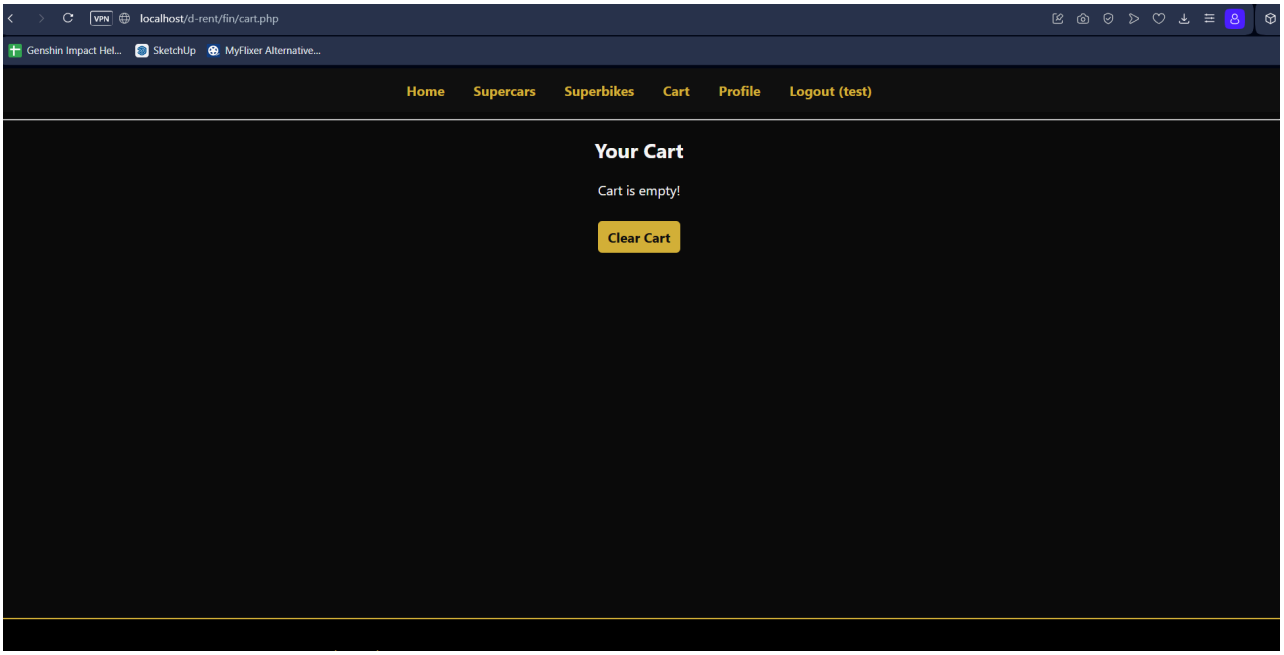


Figure 1.8 : Clear Cart Function

When proceeding to checkout, the user will be prompted to enter their full name and the number of days they wish to rent the vehicle.

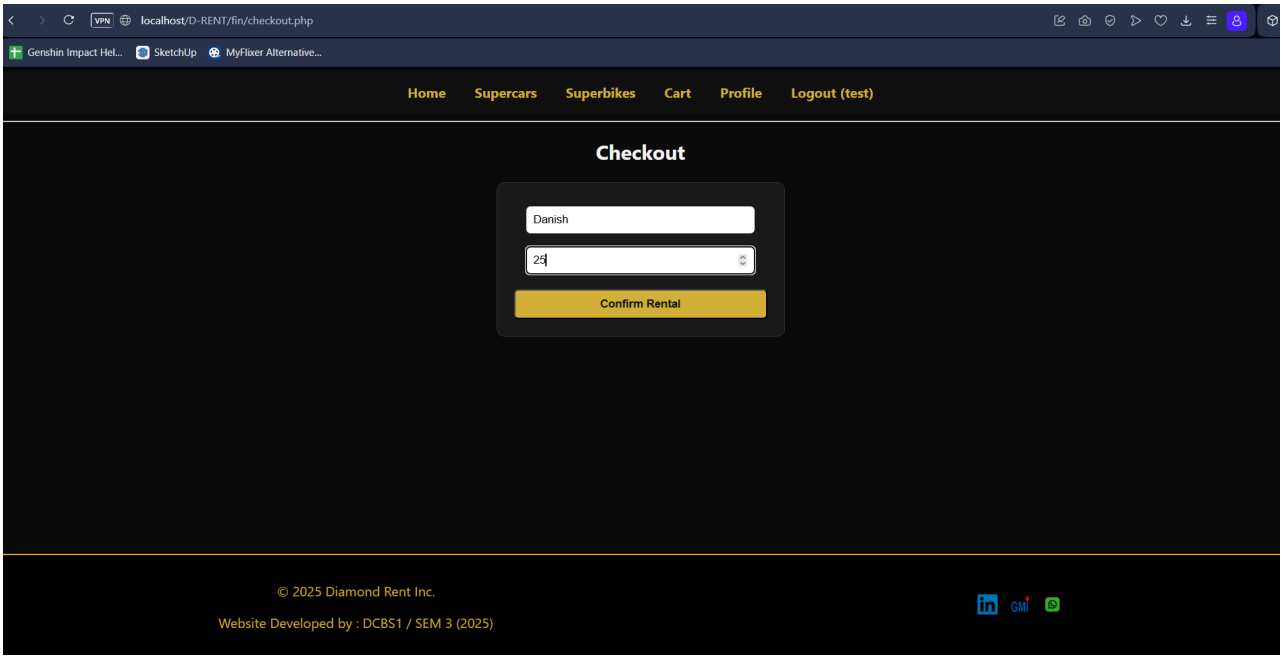


Figure 1.9 : Checkout

Upon confirming the rental, a receipt is displayed showing the rented item, the number of days, the price per day, and the total price calculated as (days × price per day).

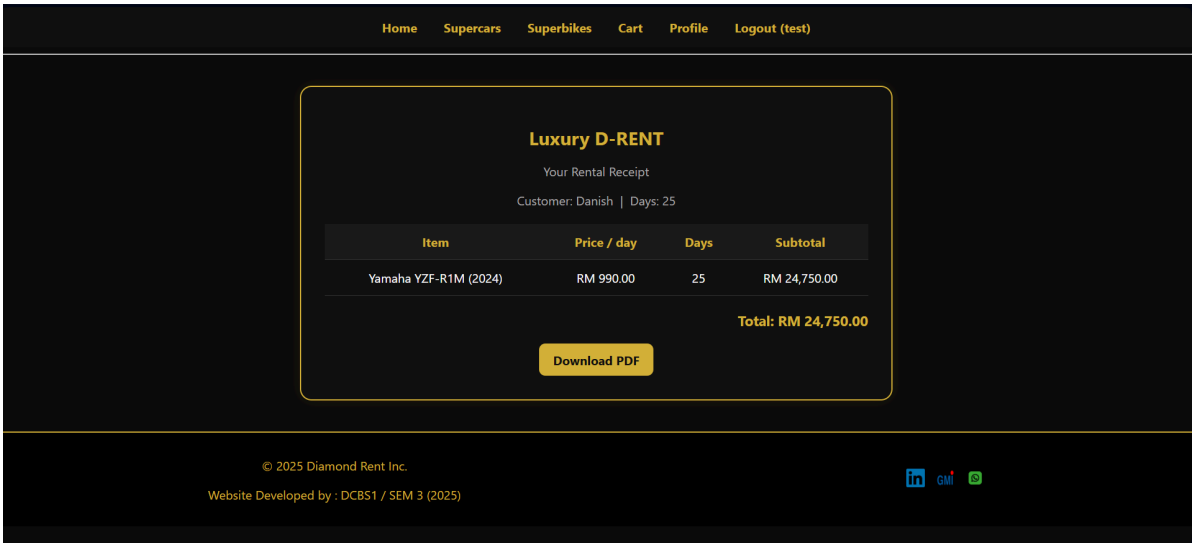
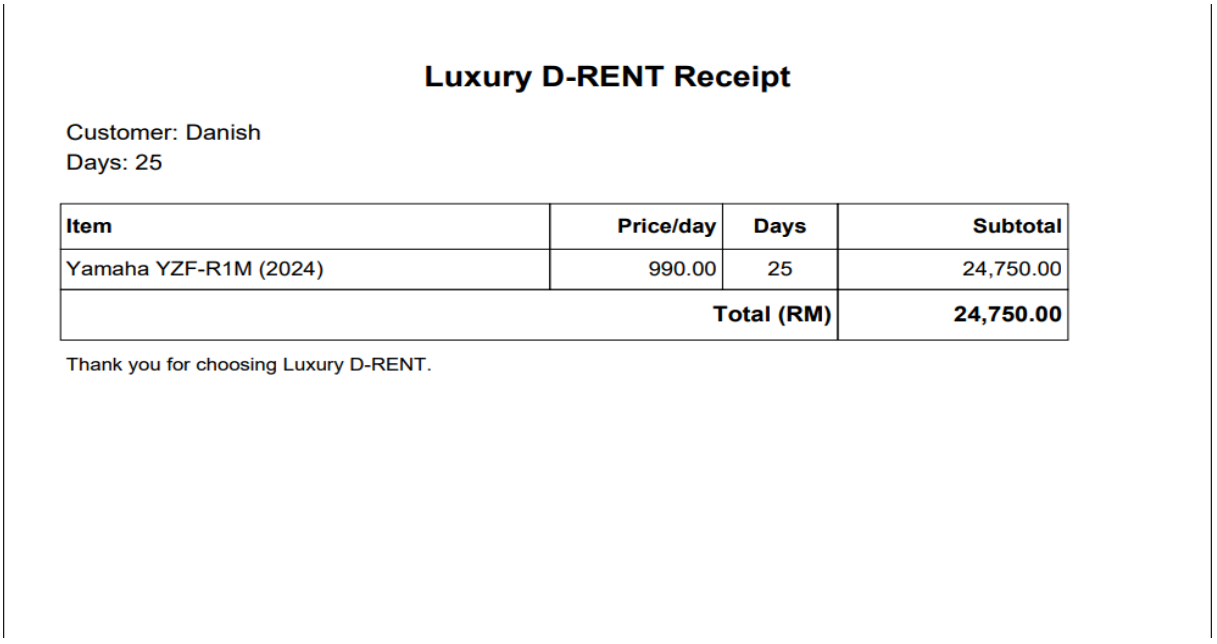


Figure 2.0 : Receipt

Clicking the 'Download PDF' button generates a receipt in PDF format, allowing the user to save it for their records.



Luxury D-RENT Receipt

Customer: Danish
Days: 25

Item	Price/day	Days	Subtotal
Yamaha YZF-R1M (2024)	990.00	25	24,750.00
		Total (RM)	**24,750.00**

Thank you for choosing Luxury D-RENT.

Figure 2.1 : Receipt to PDF

Within the profile menu, it lets users manage their personal details such as name, email, password, age and date of birth. They can choose gender, set preferences for updates, upload a driving license, rate rental satisfaction and provide details like favourite car colour, preferred car type and personal website. A **Submit** button saves the information.

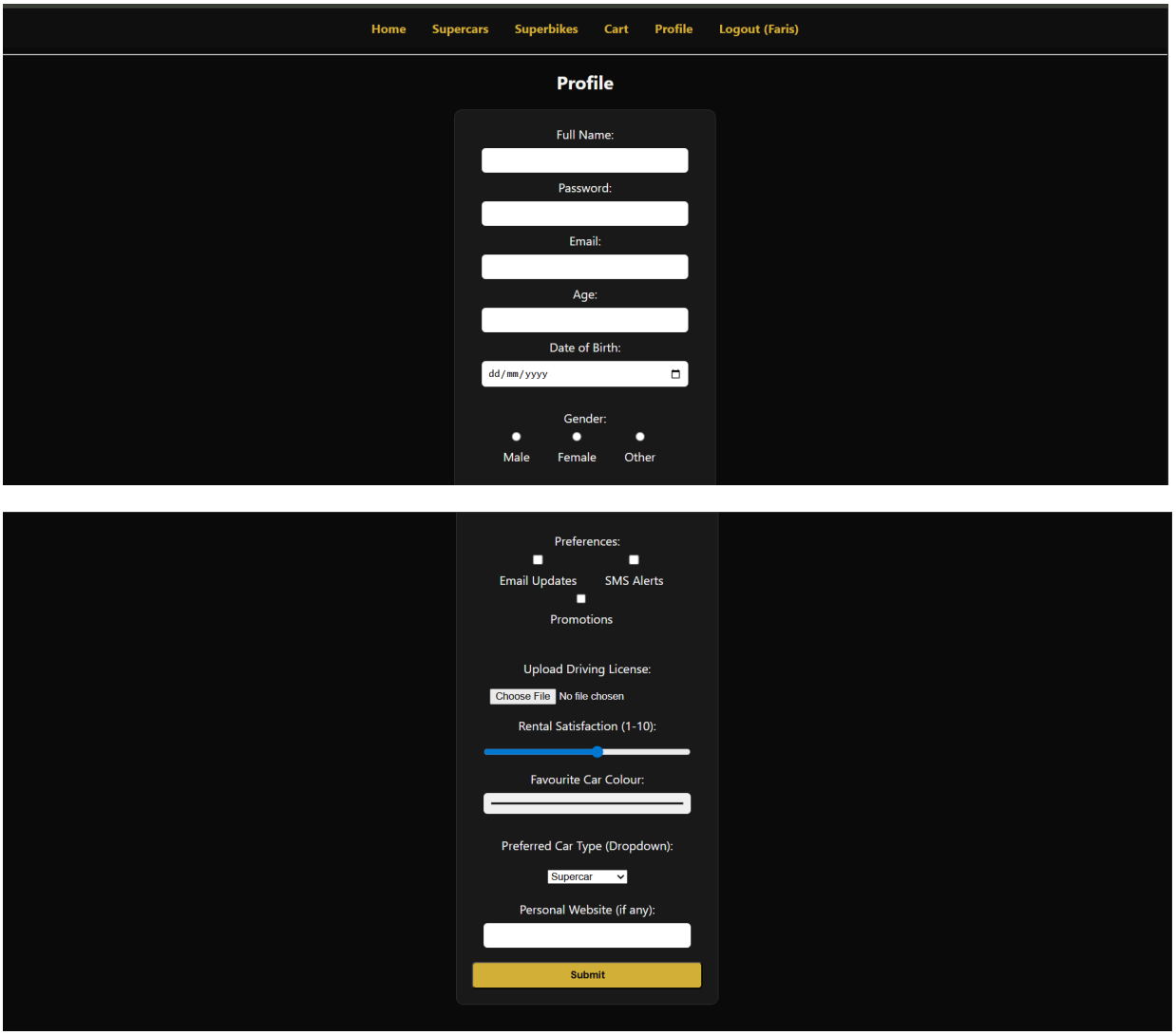


Figure 2.2 : Profile Page

Within the same menu, users can also give feedback about their opinions or suggestions by typing in the text and submitting it by "sending feedback" button.

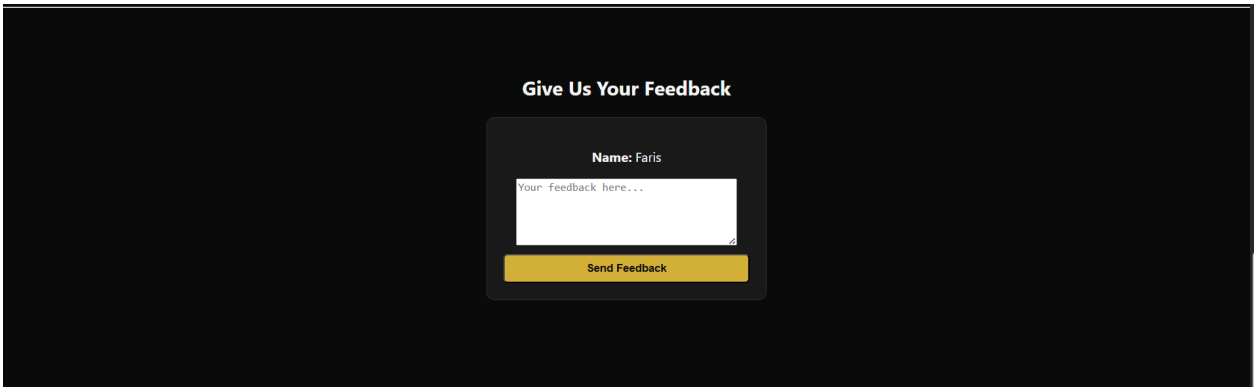


Figure 2.3 : Feedback Form

Conclusion

This project successfully achieved its goal of building an interactive and user-friendly rental website while applying the core concepts of web programming. Through the development process, we implemented secure login and registration, dynamic content presentation, and interactive features such as a shopping cart, checkout system, and PDF receipt generation. We also enhanced user engagement by including a feedback submission feature with email integration and designed a clean, professional interface using a consistent navigation bar and footer.

From a technical perspective, this assignment strengthened our understanding of HTML and PHP, as well as concepts like session handling, page routing, and external library integration (FPDF for PDF export and SMTP for email).

Overall, the system meets the specified requirements and extends beyond them by adding practical features found in real-world web applications. This exercise not only fulfilled the course learning outcomes but also improved our problem-solving and technical implementation skills, preparing us for more advanced web development and cybersecurity projects.

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